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Data Mining I

S2

NOVA-IMS 2025/2026
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Contents

Data Science

- Different Roles in Data Science

Fundamental principles of data-driven thinking

- Data-Driven Thinking: Benefits
- Data-Driven Thinking: Statistics vs Data Science
- Data-Driven Thinking: How to Build Models
- Data-Driven Thinking: The relevance of data
- Data-Driven Thinking: Building features
- The canonical tasks in Data Mining
 - Supervised learning
 - Unsupervised learning

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Data Science

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Data Science

- Data science is **the study** of where **information** comes from, what it represents and **how it can be turned into a valuable resource** in the creation of business and IT strategies.
- “Data scientists are involved with gathering data, massaging it into a tractable form, making it tell its story, and presenting that story to others,” — Mike Loukides

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Data Science vs Data Mining

• Data Science

- Data science is a **set of fundamental principles** that support and guide the principled extraction of information and knowledge from data.
- Data science **involves more than just data-mining algorithms**. Successful data scientists must be able to view business problems from a data perspective.

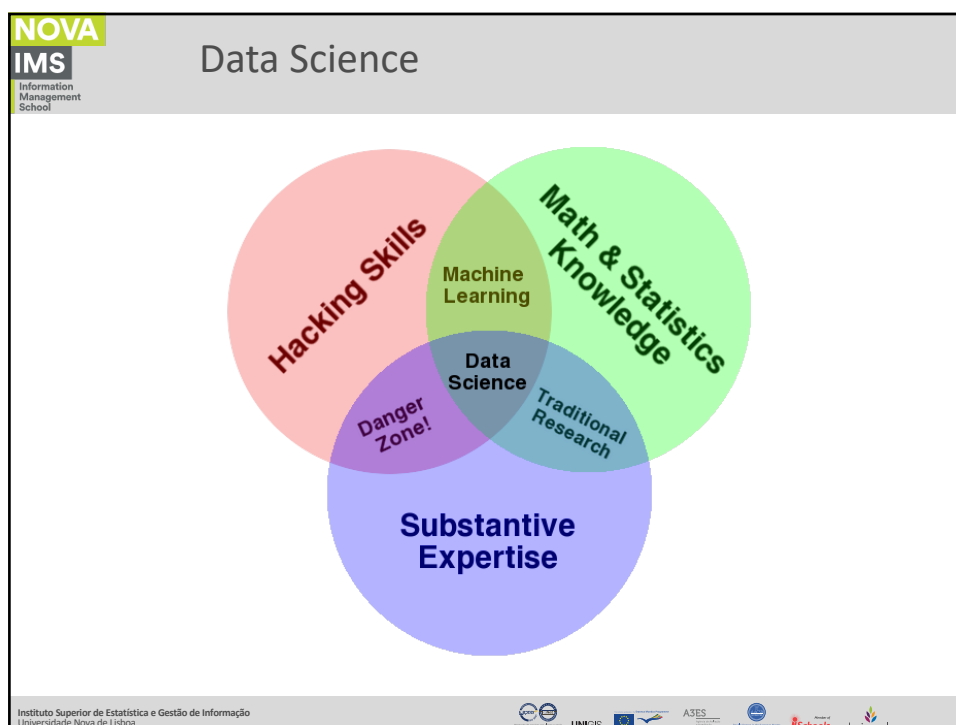
• Data mining

- The **actual extraction of knowledge** from data via technologies that incorporate the data science principles.
- There **are hundreds of different data-mining algorithms**, and a great deal of detail to the methods of the field.

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Data Science

- **iPhone Operations Data Scientist, Apple - Santa Clara Valley - California - US**
- **Key Qualifications**
 - The position requires a software programming skill set, utilization of statistical techniques, experience managing data integrity, and implementing automated solutions. The Data Scientist will need to have an **understanding of relational database management systems, design, and structured query language.**
 - Excellent analytical skills, high level of statistics with the ability to identify and predict trends and anomalies.
 - Experience in data mining **extremely large data sets, high proficiency in SQL (Teradata, Oracle, or MySQL)**, relational database management systems and design.
 - **Strong programming, with excellent object-oriented and dynamic scripting language skills.** Python, Java/C#/Objective-C, HTML5, CSS3, JavaScript, and Unix shell scripting are strongly desired.
 - Experience with statistical tools like JMP, Minitab, and Stata
 - Strong ability to manage multiple tasks concurrently and in a timely manner, including large, complex projects.
 - Effective presentation skills and be able to explain complex data and charts in a clear manner to large audiences. Outstanding communication skills, both verbal and written.

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Data Science

- **Data & Applied Scientist, Microsoft - Redmond, WA, US**
- **Job requirements:**
 - **Programming skills related to data using technologies like Python, PERL, C#, etc.**
 - Stats and data analysis experience working with **advanced tools like R, SAS and advanced Excel**
 - Experience using data to build ML and statistical models that impact a product or business
 - **Ability to access, reduce, and join large data sets programmatically**
 - Natural curiosity that extends beyond your daily activities to help stitch together different areas of the company, market and technology in new ways
 - Ability to think algorithmically about product and business issues
 - Willingness to work in a start-up organization with evolving responsibilities and a wide variety of work
 - Understanding that getting value from imperfect data and systems is a core virtue for a data scientist
 - Creativity to find pragmatic paths through ambiguity
 - Attention to detail and accuracy
 - Ability to collaborate with people representing diverse points of view
 - Solid writing, presentation and data visualization skills
 - **Other desirable skills and experiences:**
 - Engineering experience building large data systems based on SQL, Hadoop, etc.
 - Experience with product and service telemetry systems
 - Experience with Facebook/Text analytics
 - Experience conducting multivariate experiments such as A/B testing
 - Ability to bring broad product, customer, and market context into data analysis
 - Ability to connect with senior leaders to drive product and business impact
 - 2 years' experience in data exploration, analysis, programming, and modeling
 - Degree in computer science, machine learning, statistics, math, economics, business or other scientific or quant-focused field, MS or PhD preferred

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Different Roles

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Data Science – Different Roles

The diagram consists of three overlapping circles. The left circle is yellow and labeled 'Data Engineer'. The middle circle is green and labeled 'Data Scientist'. The right circle is blue and labeled 'Business Analyst'. The intersections contain specific skills or tasks.

Role	Skills/Responsibilities
Data Engineer	Coding and development Ensure quality flow, end-to-end
Data Scientist	Statistics Improve algorithm models to boost bottom-line
Business Analyst	Customer-related data Cross-department analysis

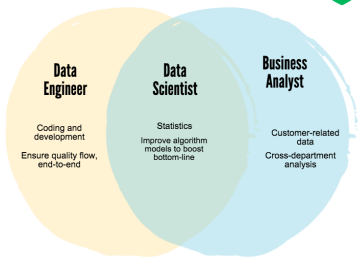
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Data Science – Different Roles



The Venn diagram illustrates the intersection of skills for three roles: Data Engineer (yellow circle), Data Scientist (green circle), and Business Analyst (blue circle). The intersections show shared skills: Data Engineer and Data Scientist share 'Coding and development' and 'Ensure quality flow, end-to-end'; Data Scientist and Business Analyst share 'Statistics' and 'Improve algorithm models to boost bottom-line'; Data Engineer and Business Analyst share 'Customer-related data' and 'Cross-department analysis'. The center intersection of all three roles is labeled 'H'.

- **Business Analyst:**
 - Business analysts' strengths lie in their business acumen.
 - They can communicate well with both the data scientist and C-suite to help drive data-driven decisions faster.
 - The best business analysts also have skills in statistics to be able to glean interesting insights from past behavior.

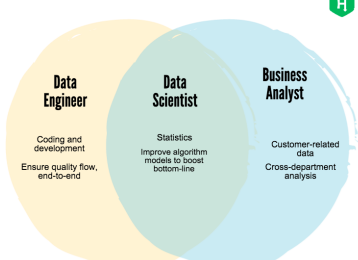
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Data Science – Different Roles



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- **Data Scientist:**
 - Data science is largely rooted in statistics, data modeling, analytics and algorithms.
 - They focus on conducting research, optimizing data to help companies get better at what they do.
 - The minds behind recommended products on Amazon.

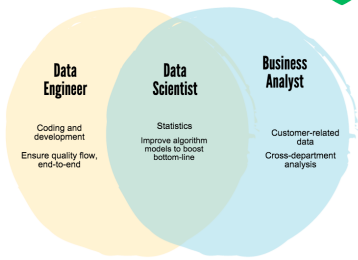
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Data Science – Different Roles



The diagram consists of three overlapping circles. The left circle is yellow and labeled 'Data Engineer' with sub-points: 'Coding and development' and 'Ensure quality flow, end-to-end'. The middle circle is green and labeled 'Data Scientist' with sub-points: 'Statistics' and 'Improve algorithm models to boost bottom-line'. The right circle is blue and labeled 'Business Analyst' with sub-points: 'Customer-related data' and 'Cross-department analysis'. A small green hexagon with a white 'H' is positioned above the right circle.

- **Data Engineer:**
 - While data scientists dig into the research and visualization of data, data engineers ensure the data is powered and flows correctly through the pipeline.
 - They're typically software engineers who can engineer a strong foundation for data scientists or analysts to think critically about the data.

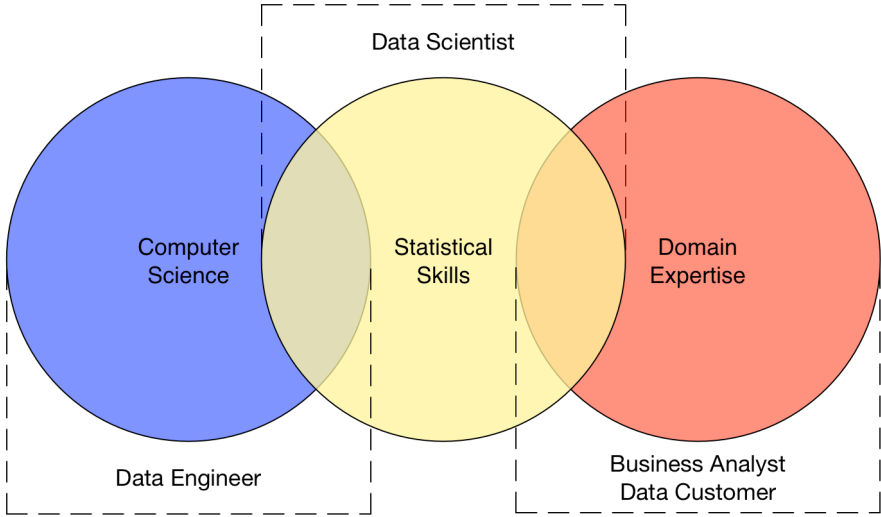
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Data Science – Different Roles



The diagram consists of three overlapping circles. The left circle is blue and labeled 'Computer Science'. The middle circle is yellow and labeled 'Statistical Skills'. The right circle is red and labeled 'Domain Expertise'. Dashed boxes group the circles into three categories: 'Data Engineer' (Computer Science and Statistical Skills), 'Data Scientist' (Statistical Skills and Domain Expertise), and 'Business Analyst Data Customer' (Domain Expertise).

Fonte: <https://towardsdatascience.com/data-engineer-vs-data-scientist-vs-business-analyst-b68d201364bc>

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Data-Driven Thinking Benefits

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Data-Driven Thinking

*“There will be a **shortage of talent** necessary for organizations to take advantage of big data.*

*By 2018, the United States alone could face a shortage of **140,000 to 190,000 people with deep analytical skills** as well as **1.5 million managers and analysts with the know-how to use the analysis** of big data to make effective decisions.”*

Fonte: McKinsey Big data: The next frontier for innovation, competition, and productivity

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Data-Driven Thinking

Data scientist

you just lost The Game by 9GAG.COM

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Data-Driven Thinking

Statistics vs Data Science

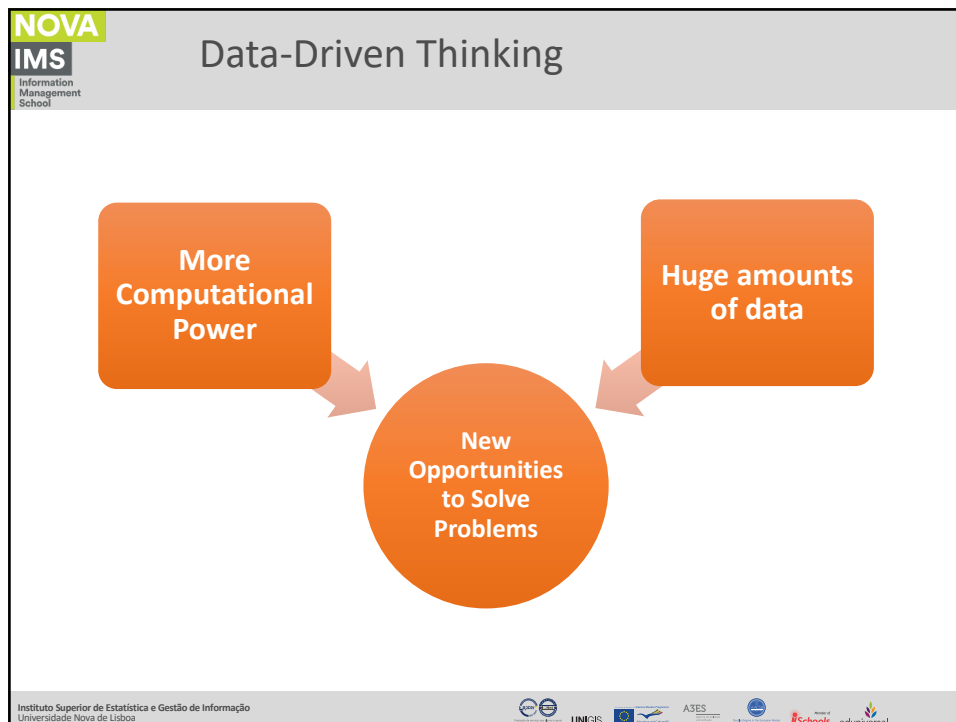
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What new about Data Mining?

(Data Mining: Statistics and More? David J. Hand)

- Statistics might be described as being characterized by **data sets which are small and clean**, which are **static**, which were **sampled in an iid manner**, which were often **collected to answer the particular problem** being addressed, and which are **solely numeric**.
- None of these apply in the data mining context.....

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What is Data Mining?

	Experimental Primary	Opportunistic Secondary
Purpose	Research	Operational
Value	Scientific	Commercial
Origin	Controlled	Passively observed
Size	Small	Massive
Hygiene	Clean	Dirty
Status	Static	Dynamic

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Data-Driven Thinking How to Build Models

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Find/Build Attributes



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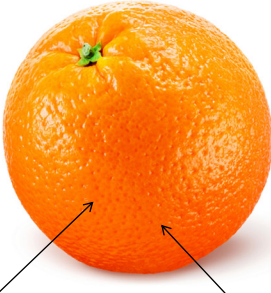
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Find/Build Attributes

- **Features/Attributes**
 - **Features** are fundamental to train an ML system
 - They are the **properties of the things you're trying to learn about**



Weight: 340g

Colour: Orange

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Find/Build Attributes

- Features/Attributes**
 - Features of a fruit might be **weight and colour**. 2 features would mean 2 dimensions
 - 2 dimensions can be plotted in a graphic provided they are expressed numerically

Plotting apples (red and green dots) and oranges (orange dots) weight and colour

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Find/Build Attributes

- Features/Attributes**
 - With these features the **ML system can learn** to split data up with a **line to separate oranges from apples**
 - This can now be **used to make future classifications** when we plot new points the system has not seen

Plotting apples (red and green dots) and oranges (orange dots) weight and colour

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Find/Build Attributes

- **Features/Attributes**
 - An ML system cannot predict about stuff it does not know about
 - Classify a papaya
 - This is because it only knows about apples and oranges and this was the closest match

Plotting apples (red and green dots) and oranges (orange dots) weight and colour

Weight

Colour

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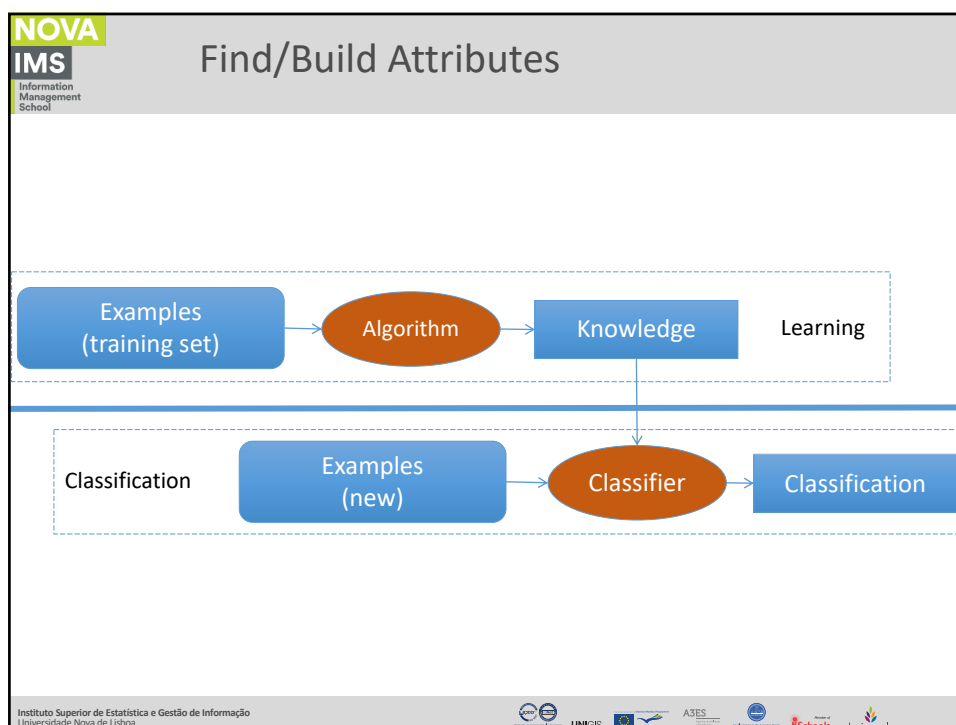
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Let's build a model!!!

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Data-Driven Thinking The Relevance of Data

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Find/Build Attributes

- **Features/Attributes**
 - Choosing the **appropriate features** has a major impact on the performance of any ML system
 - **Some features will never allow** the system to produce **good results**
 - How to choose? Practice and **knowledge about the problem**

Plotting apples and oranges ripeness and number of seeds

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Find/Build Attributes

More features = higher probability of discriminating (up to a point)

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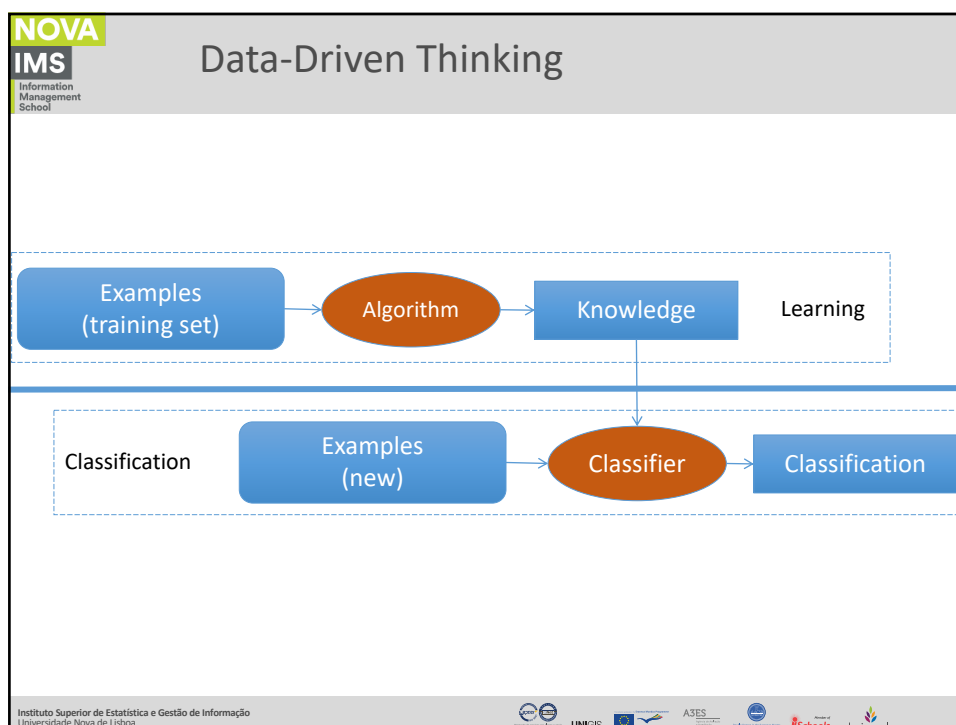
Find/Build Attributes



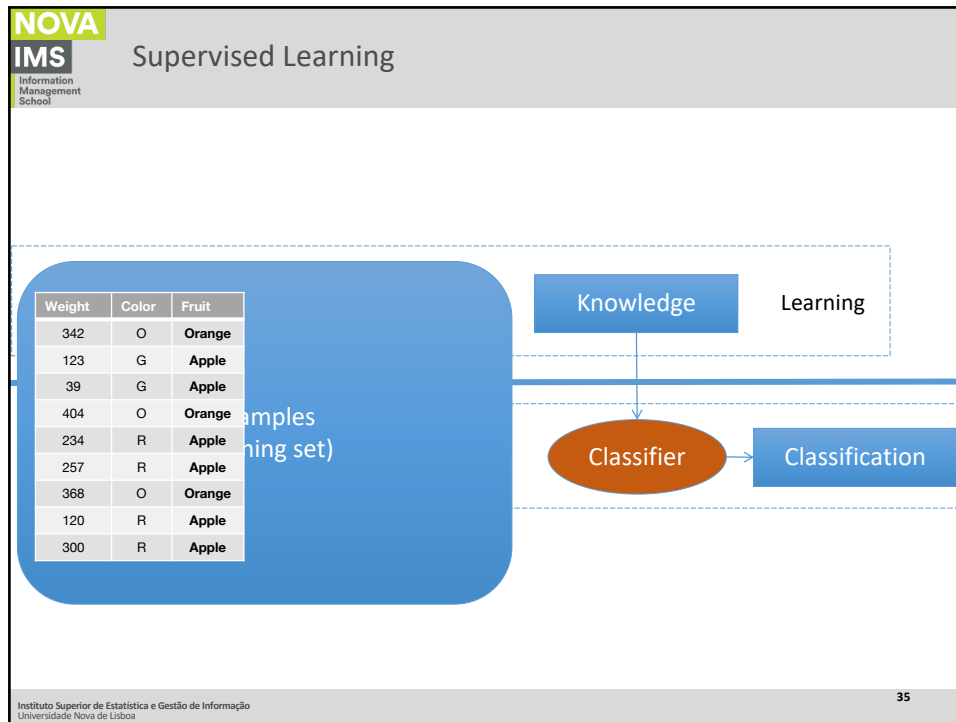
- Do you know **which examples correspond to apples** and which correspond to oranges?
 - We need the labels (fraud)
- Do you have **enough labeled examples**?
 - We need experience (scarce)
- Do you know **what an orange is**?
 - We need clear cut definitions (churn)

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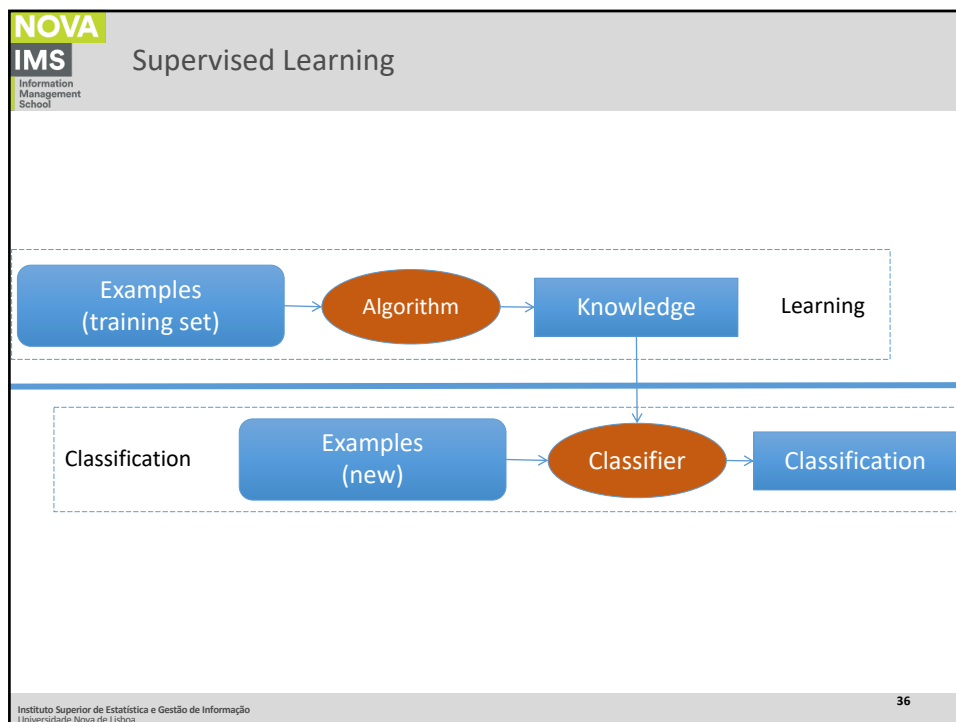
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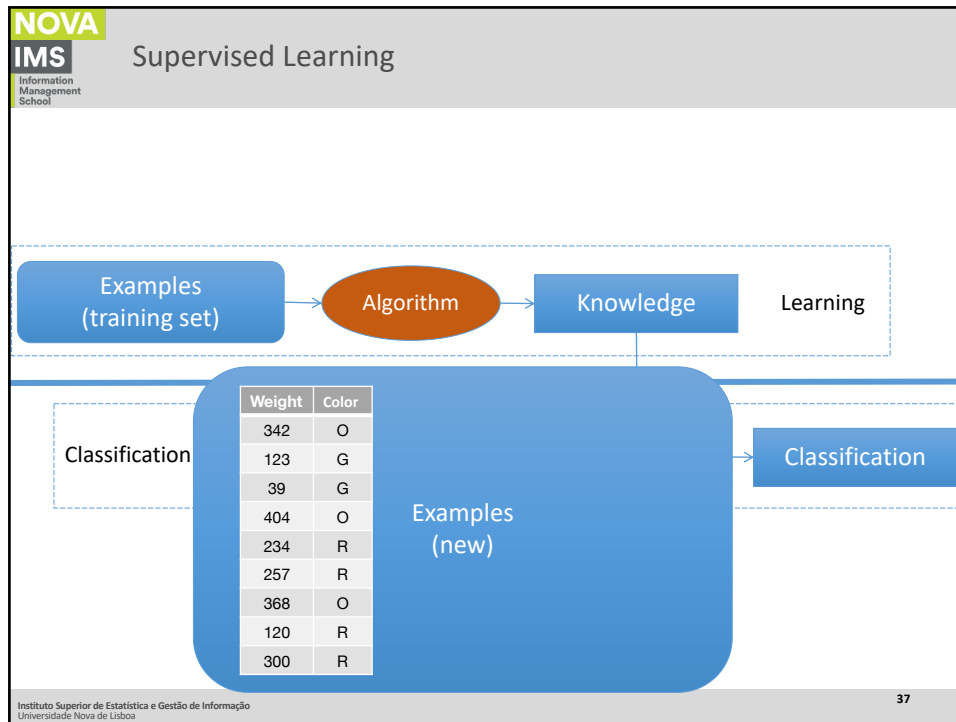
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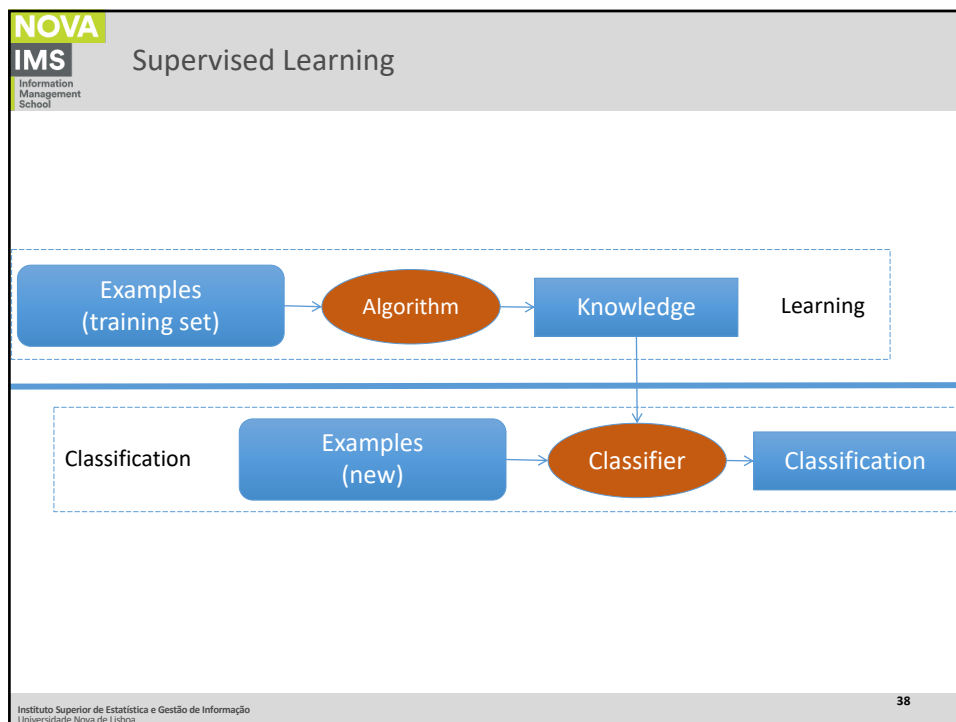
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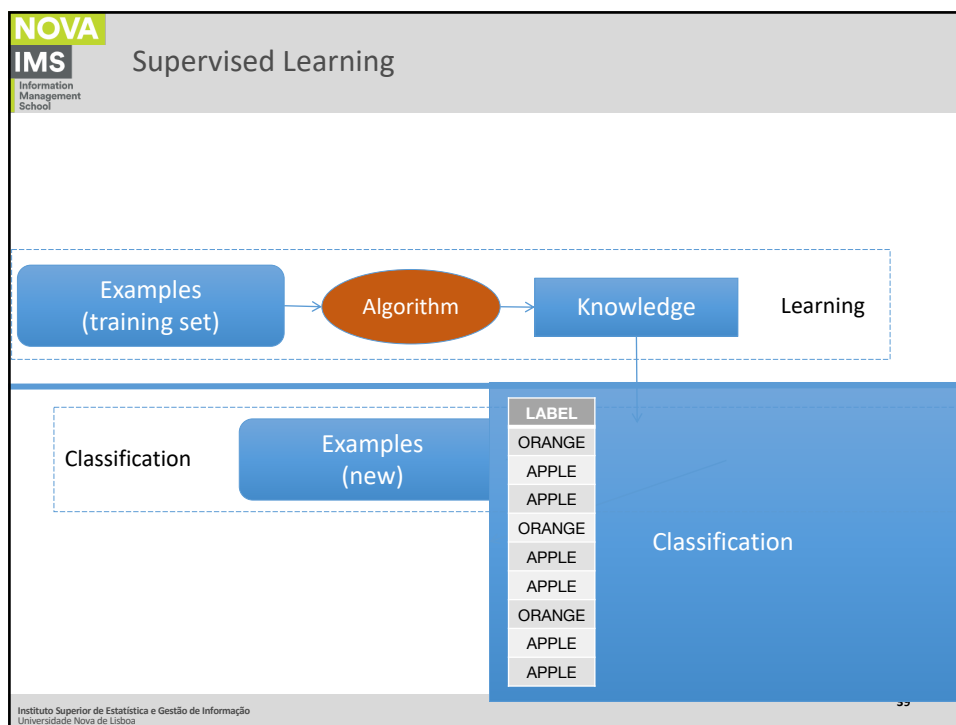
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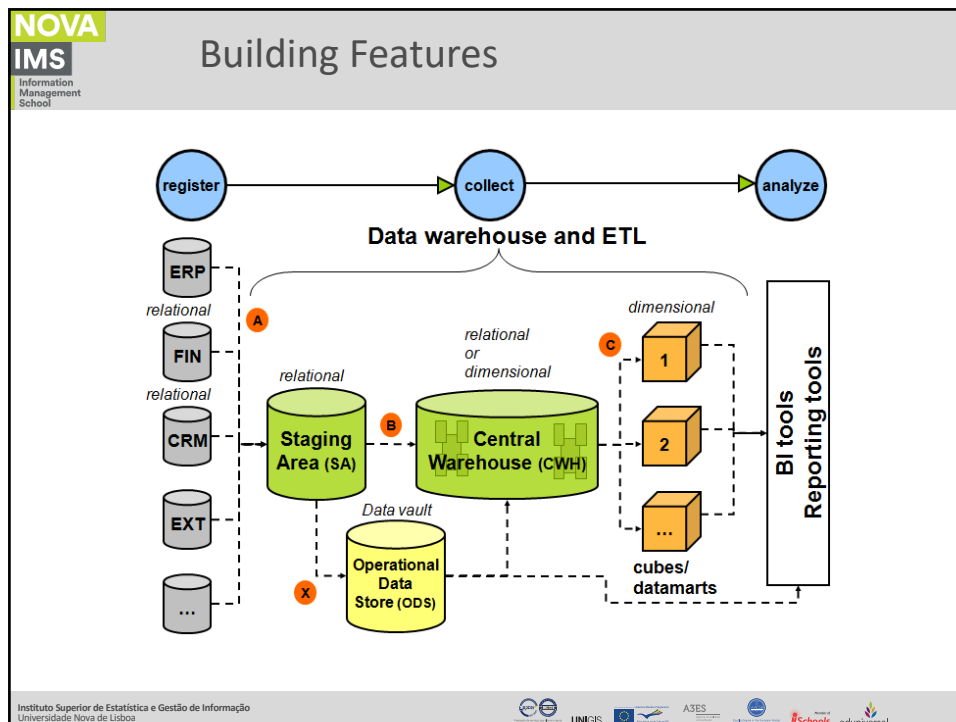
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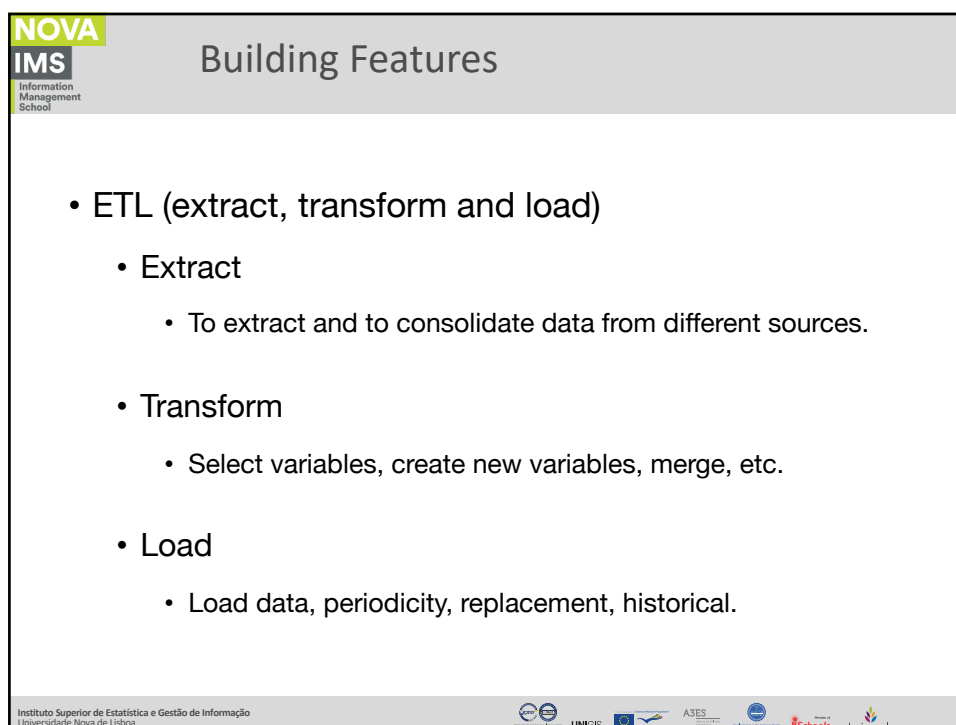
Data-Driven Thinking Building Features

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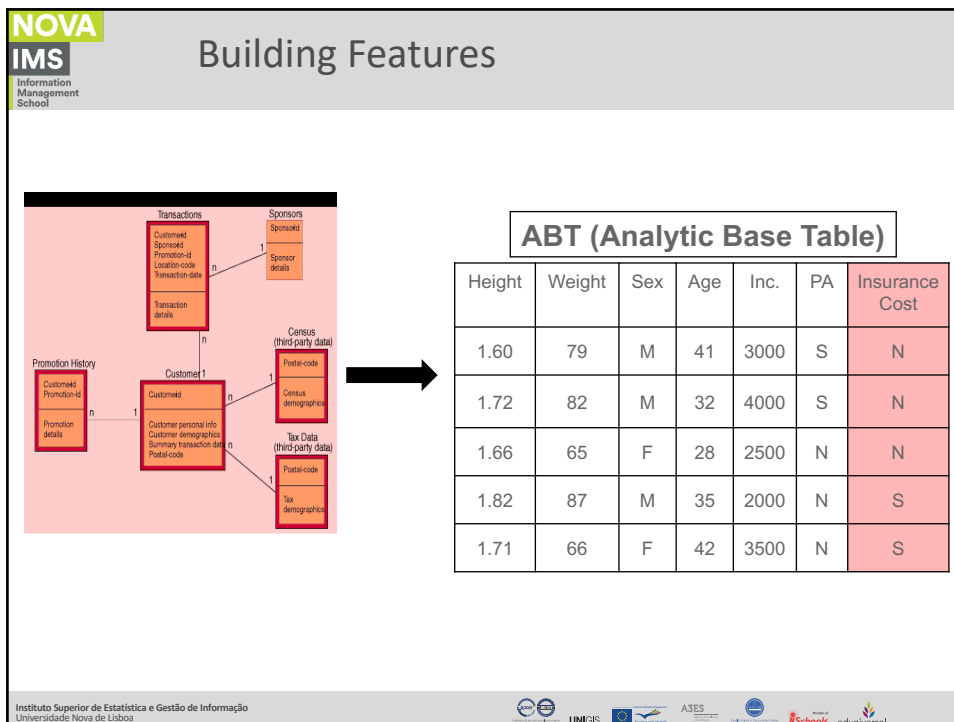
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Let's build features!!!

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Find/Build Attributes

The minimum amount of information required to identify customer behavior is specified in the following list:

1. Transaction number
2. Date and time of transaction
3. Item purchased (identified by UPC code or equivalent)
4. Product price (per item or by unit of measurement)
5. Quantity purchased (number of items or units purchased)
6. A table matching product code to product name, subgroup code to subgroup name, and product group code to product group name
7. A product taxonomy that links product code to product subgroup code and product subgroup code to product group code.

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- **Some relevant variable**
 - **Recency** – day since last visit/purchase
 - **Frequency** – number of transactions per customer
 - **Monetary Value** – total value of sales (different from profit)
 - **Average Purchase** – average of the purchase per visit
 - **Most Frequent Store**
 - **Average Time Between Transactions** – Transaction Interval
 - **Standard Deviation of Transactional Interval**
 - **Customer Stability Index** - Standard Deviation of Transactional Interval/Average Time Between Transactions
 - **Relative Spend on Each Product**

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The Canonical Tasks in Data Mining

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Different Tasks

```

graph LR
    DM[Data Mining] --> PM[Predictive modelling*]
    DM --> DM2[Descriptive modeling**]
    PM --> PM_desc[To learn a decision criterion that allows us to classify new and unknown examples;]
    DM2 --> DM2_desc[To describe, to provide a summary of a data set;]
  
```

* aka supervised learning
** aka unsupervised learning

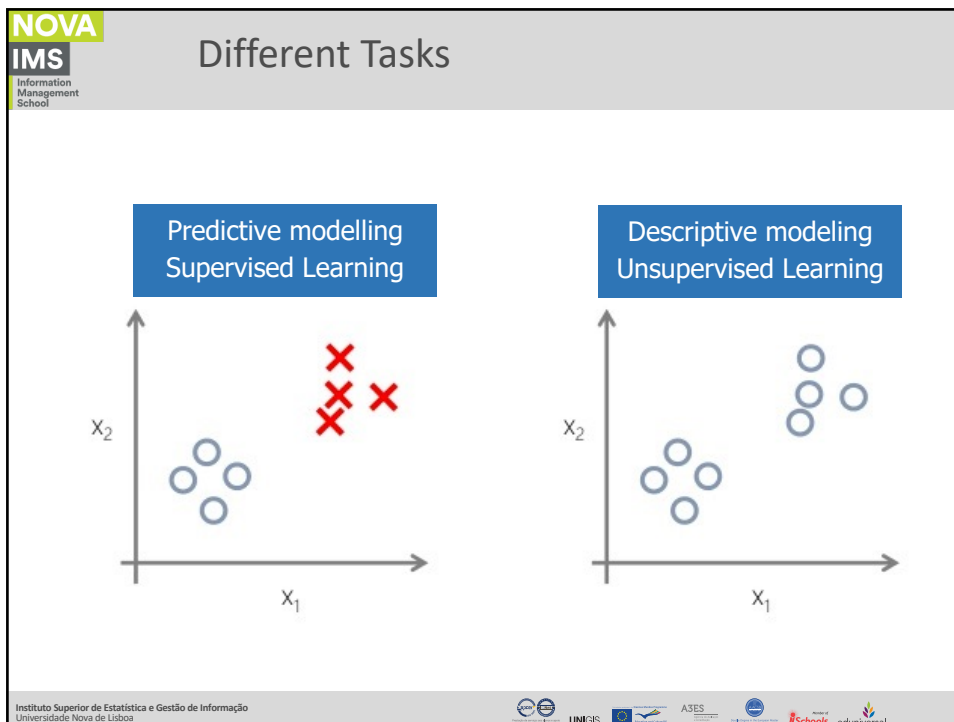
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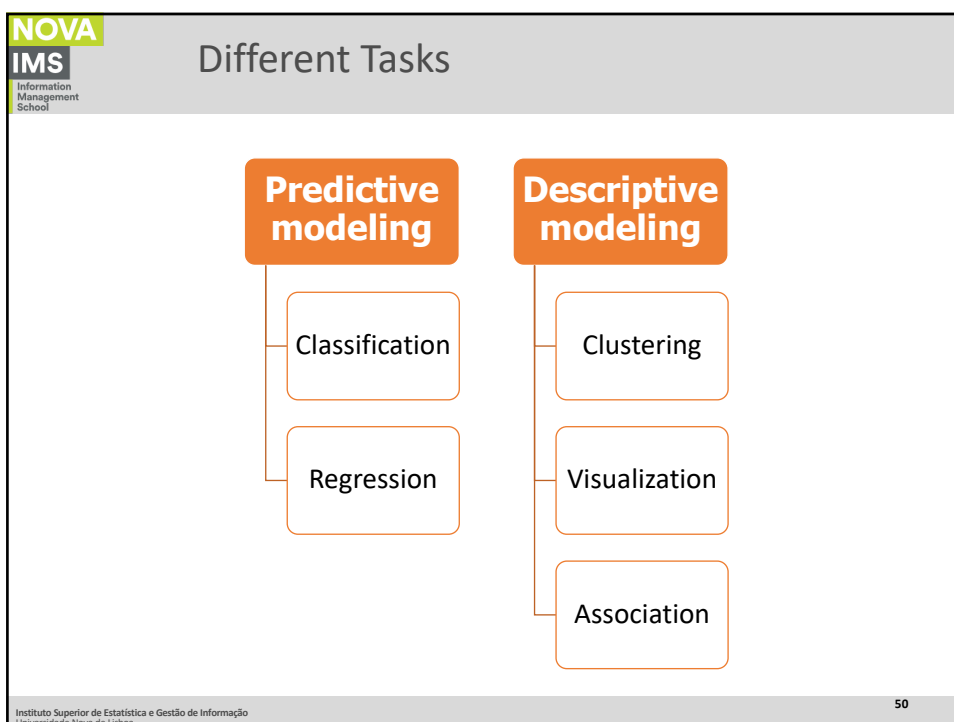
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```
graph LR; DM[Data Mining] --- PM[Predictive modelling*]; DM --- DM2[Descriptive modeling**]; PM --- PM1[To learn a decision criterion that allows us to classify new and unknown examples;]; DM2 --- DM3[To describe, to provide a summary of a data set;]
```

The diagram illustrates the components of Data Mining. A central box labeled "Data Mining" is connected to two sub-tasks: "Predictive modelling*" and "Descriptive modeling**". Each sub-task is further detailed with a description of its goal.

- Data Mining**
 - Predictive modelling***
 - To learn a decision criterion that allows us to classify new and unknown examples;
 - Descriptive modeling****
 - To describe, to provide a summary of a data set;

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Predictive Modeling

Feature

Label

Height	Weight	Sex	Age	Income	Physical Activity	Insurance Costs
1.60	79	M	41	3000	S	N
1.72	82	M	32	4000	S	N
1.66	65	F	28	2500	N	N
1.82	87	M	35	2000	N	S
1.71	66	F	42	3500	N	S

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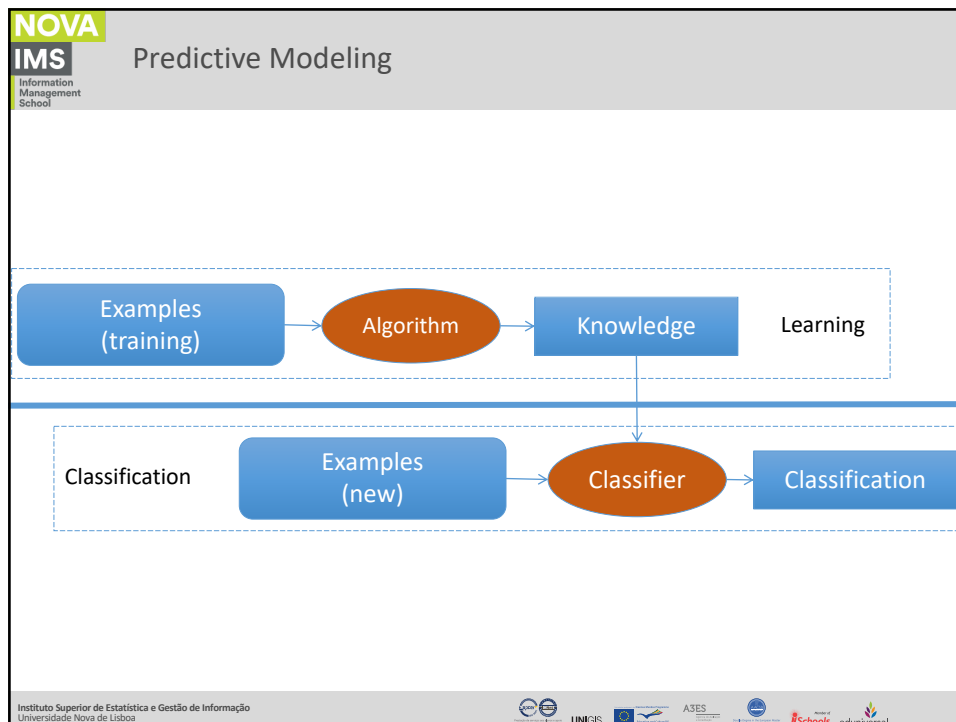
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- A real estate agency wants to estimate the price range for each customer based on their income;
- Training examples:
 - Historical data;
 - Income vs sold house prices.

House price

Income

```

graph LR
    subgraph Learning
        A[Examples (training)] --> B(Algorithm)
        B --> C[Knowledge]
    end
    subgraph Classification
        D[Examples (new)] --> E(Classifier)
        E --> F[Classification]
    end
    C --> E
  
```

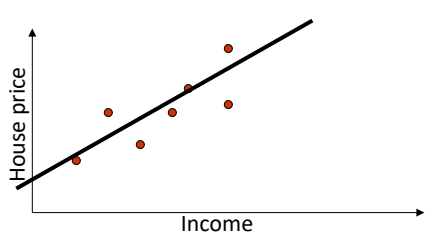
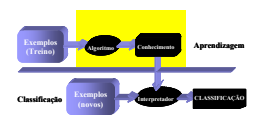
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Predictive Modeling

- Algorithm
 - Linear regression
- Knowledge representation
 - Regression line (slope and origin)

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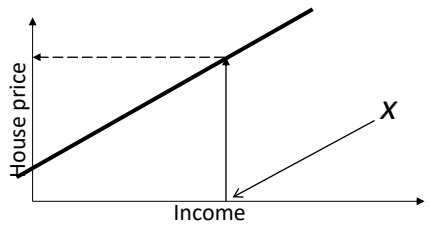
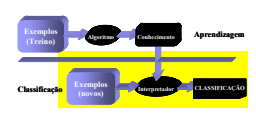
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Predictive Modeling

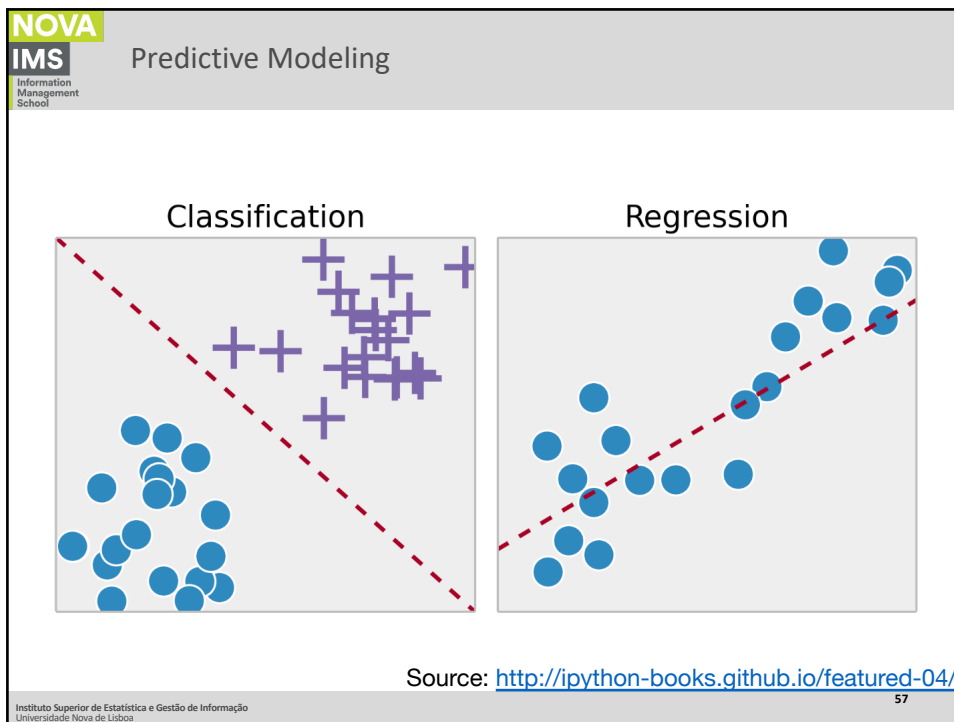
- New examples
 - A customer with an income of x
- Interpretation
 - Use the line (prediction method) to obtain an estimate

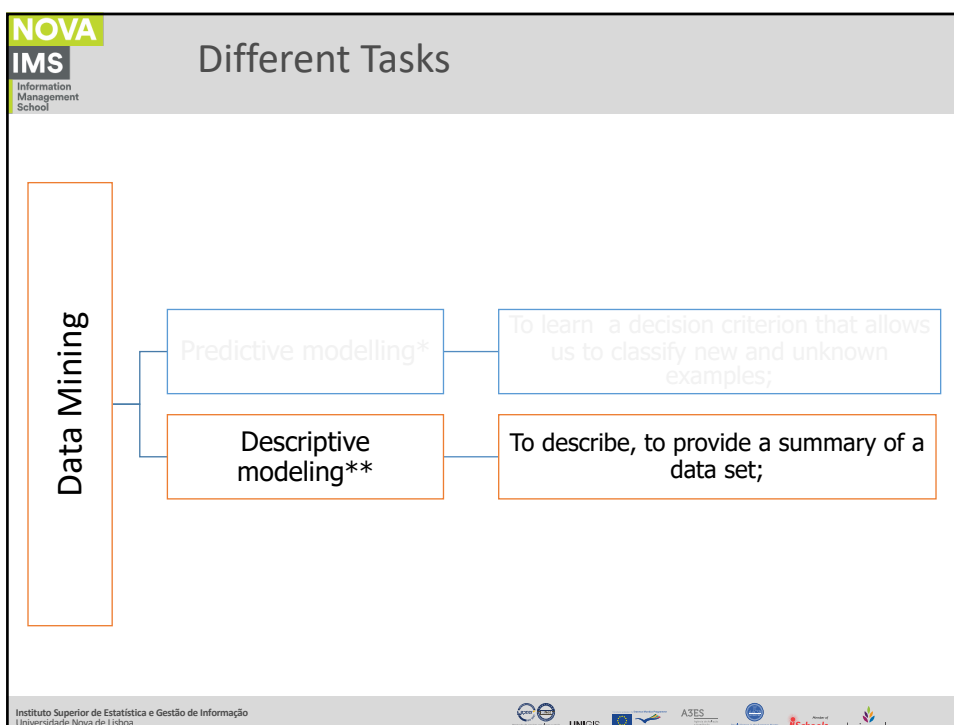
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Different Tasks

Feature 

Height	Weight	Sex	Age	Income	Physical Activity
1.60	79	M	41	3000	S
1.72	82	M	32	4000	S
1.66	65	F	28	2500	N
1.82	87	M	35	2000	N
1.71	66	F	42	3500	N

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Clustering



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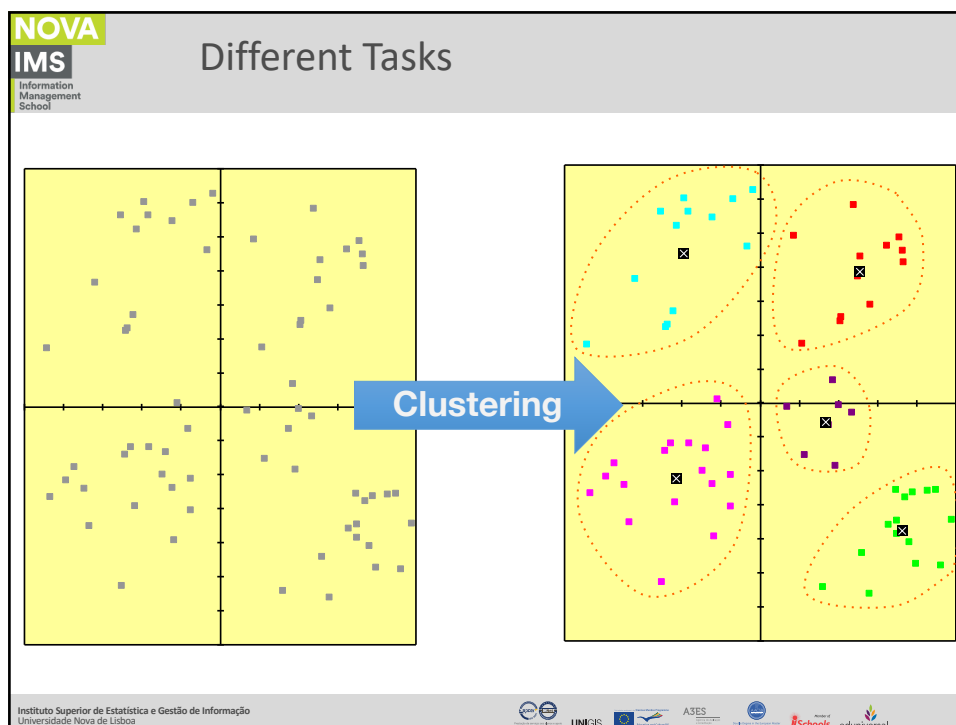




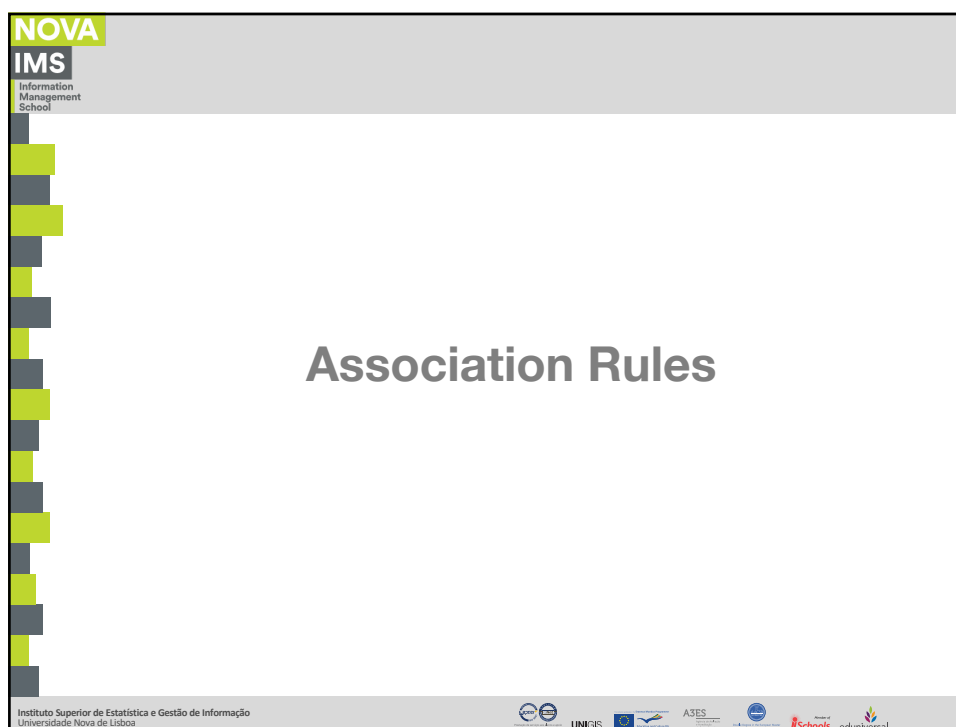




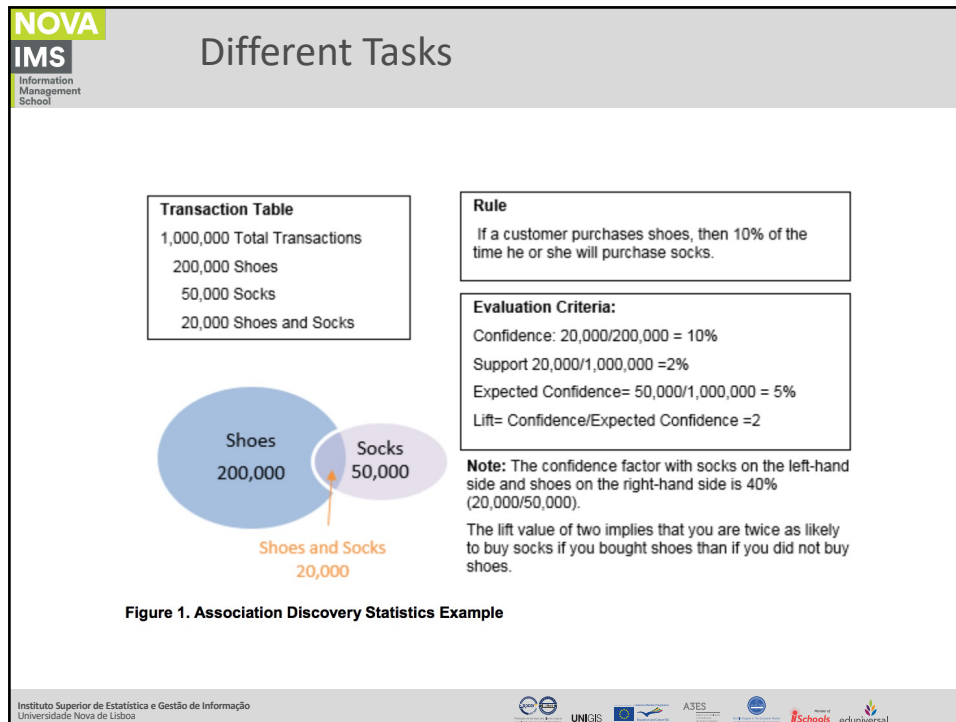

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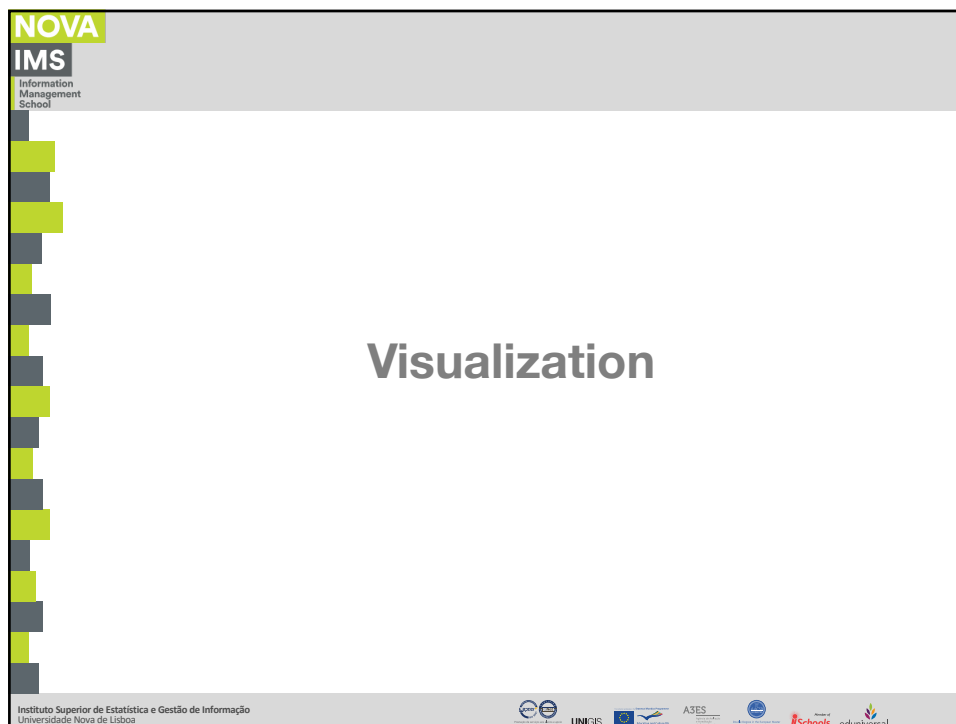
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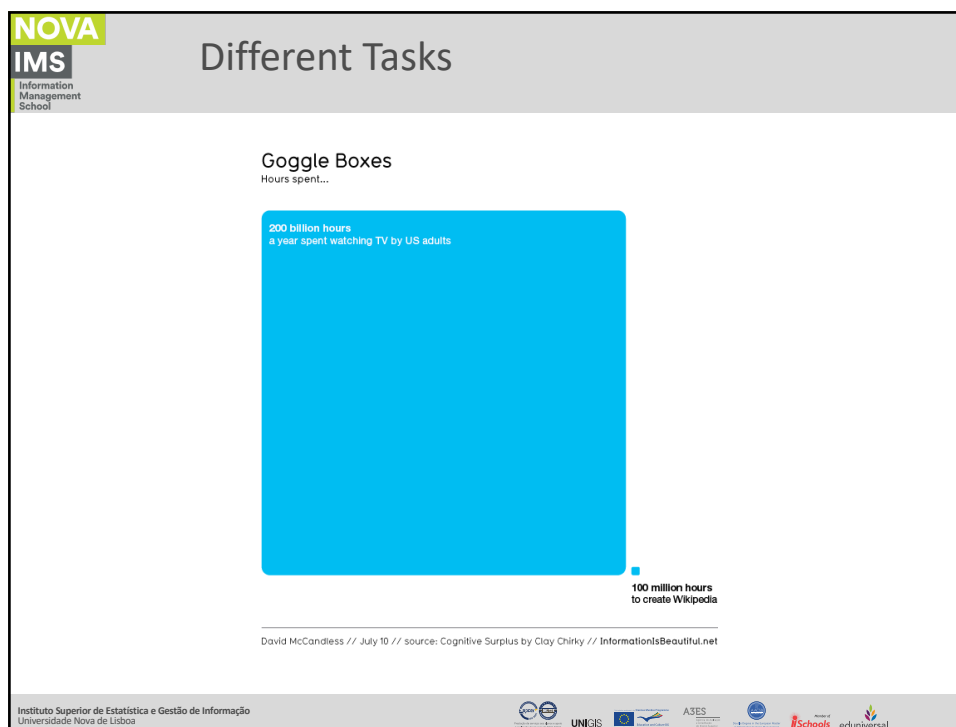
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